

Ramanand Tomar

Blockchain Developer — Backend Engineer — Systems Architect

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in **Blockchain systems, backend engineering, and distributed architectures**. Built decentralized healthcare platforms and high-performance proxy servers applying real-world **DSA, system design with Microservices Architecture, and DevOps concepts**. Finalist in national hackathons with ability to deliver end-to-end technical solutions.

TECHNICAL SKILLS

Blockchain: Ethereum, Solidity, Smart Contracts, Web3.js, MetaMask, IPFS, Truffle, Ganache, Solana (Learning)
Backend: Node.js, Express.js, REST APIs, JWT, Microservices, Kafka, RabbitMQ, Clustering, Caching
DevOps: Docker, Jenkins, Kubernetes, Prometheus, Grafana, Ansible, Terraform, AWS (EC2, S3)
Frontend: React.js, Tailwind CSS, HTML5, JavaScript (ES6+), Context API
Databases: MongoDB, MySQL, Database Design, Query Optimization
Languages: JavaScript, C++, Python, Rust, SQL, Data Structures & Algorithms
Tools: Git, GitHub, Vercel, Render, Postman, Linux/Unix Command Line

TECHNICAL PROJECTS

ArogyaBridge — Blockchain Based Decentralized Healthcare System **Finalist - HackOrbit 2025**
Ethereum — Solidity — React — Node.js — IPFS — MongoDB [Live Demo](#) — [GitHub](#)

- Designed and developed a decentralized healthcare platform with MetaMask-based authentication for patients, doctors, and diagnostic centers.
- Implemented permissioned access control using Ethereum smart contracts with immutable audit logs.
- Integrated IPFS-based decentralized storage for secure medical records and auto-generated PDF reports.
- Built AI-powered medical records summarization and voice-based assistant using Gemini API.
- Developed real-time analytics dashboard with role-based admin access following MVC architecture.
- Deployed full-stack application using cloud platforms with production-ready configurations.

Multi-threaded HTTP Proxy Server with LRU Cache **Systems Project**
Node.js — Distributed Systems — Data Structures — Networking [GitHub](#)

- Designed a concurrent HTTP proxy system using **master-worker architecture** with Node.js cluster module to handle high throughput requests.
- Applied **Data Structures design** by implementing an LRU cache using **Hash Map + Doubly Linked List** achieving $O(1)$ get/put operations.
- Optimized system performance by reducing average response time through **cache-aware request routing** and memory-efficient eviction policies.
- Engineered **inter-process communication (IPC)** mechanisms for coordination between master and worker processes enabling load balancing.
- Addressed **scalability and fault tolerance** by dynamically managing worker lifecycles and isolating failures.
- Designed system-level monitoring to analyze **cache hit ratio, latency, memory usage, and worker utilization** for performance tuning.

Blue Notes — Decentralized Notes DApp on Solana **Web3 Project**
Next.js — TypeScript — Solana — Anchor — Tailwind CSS [GitHub](#)

- Built a fully decentralized, on-chain notes application where users can **create, update, and delete notes** secured and verified by the Solana blockchain.
- Integrated Phantom wallet for seamless wallet-based authentication and transaction signing without any central back-end infrastructure.
- Implemented Solana Program (smart contract) using **Rust and Anchor framework** for on-chain note storage, ensuring immutability and transparent ownership.
- Employed **Next.js + TypeScript** for responsive frontend UI with real-time updates after SOL transactions.
- Utilized Solana PDAs (Program Derived Addresses) for unique note identification and secure author verification.
- Designed UI with Tailwind CSS and wallet adapters for smooth UX across devices.

EDUCATION

Madhav Institute of Technology and Science, Gwalior

July 2023 – June 2027

Bachelor of Technology in Computer Science Engineering

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Computer Networks

ACHIEVEMENTS

- **Finalist** — HackOrbit National Level Hackathon 2025 (Top 10% among 500+ teams)
- **Finalist** — All India Developers Challenge (AIDC) 2025
- Currently learning **Solana blockchain development** using Anchor framework and Rust
- Exploring **cloud-native DevOps tools** and CI/CD pipelines with Kubernetes and Terraform

ADDITIONAL INFORMATION

Open to internship and full-time roles in Blockchain, Backend, Systems, and DevOps Engineering. Strong problem-solving skills with real-world application of data structures and system design principles.